

DATE : 07.07.2025

DT/NT : DT

LESSON : DEEP LEARNING

**SUBJECT: RECURRENT NEURAL
NETWORK – LSTM – GRU**

BATCH : 287

**DATA
SCIENCE**



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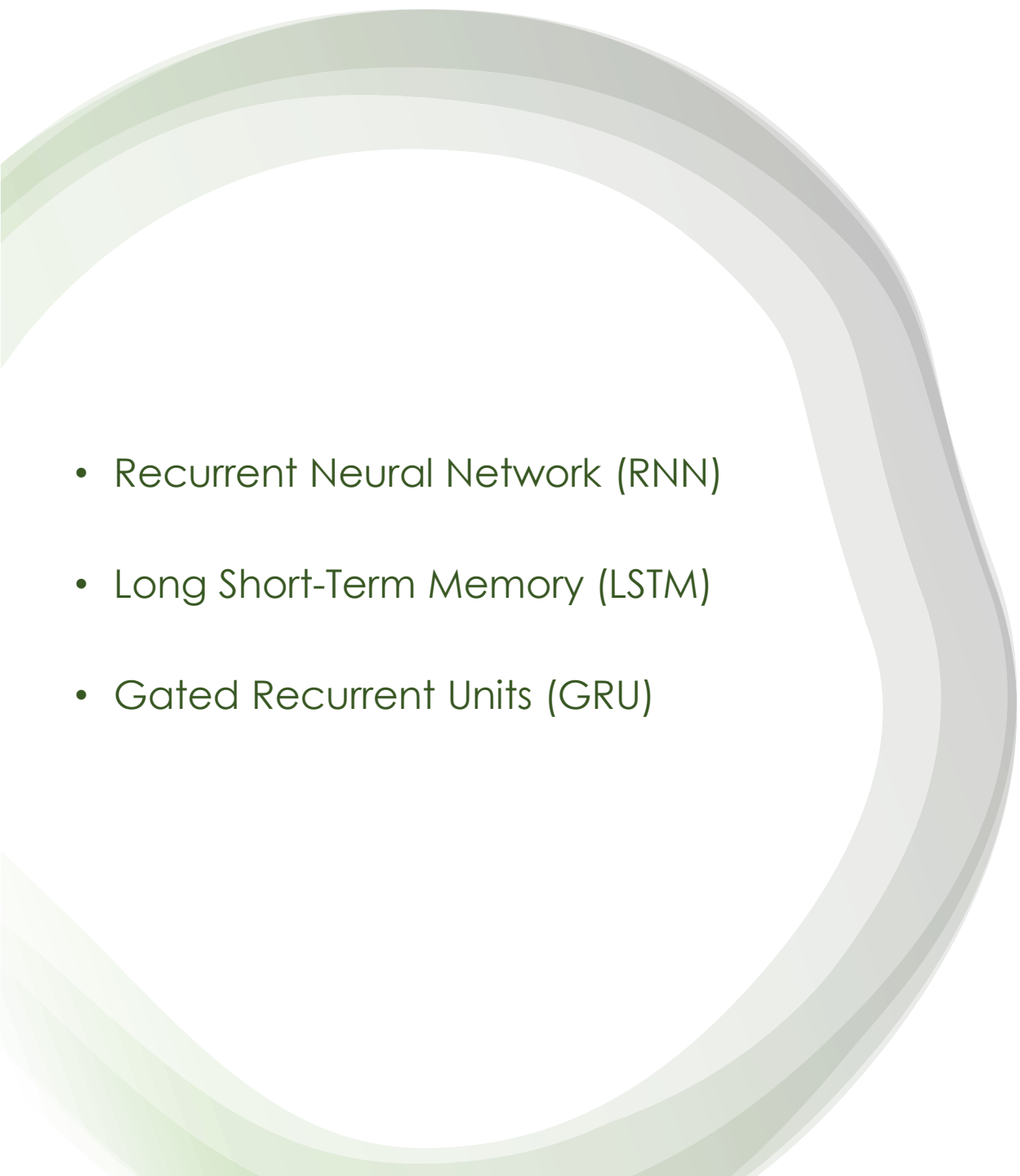


The background features abstract, thick, curved shapes in shades of green and grey. On the left, a green shape curves upwards. On the right, a large grey shape curves downwards, resembling a stylized 'C' or a partial circle. The text is positioned in the center-left area.

DEEP LEARNING II

Recurrent Neural Network (RNN)



- 
- Recurrent Neural Network (RNN)
 - Long Short-Term Memory (LSTM)
 - Gated Recurrent Units (GRU)

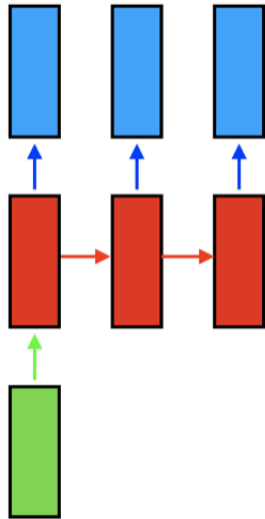
GENERAL CONTENTS

Different Types of Sequence Modeling Tasks

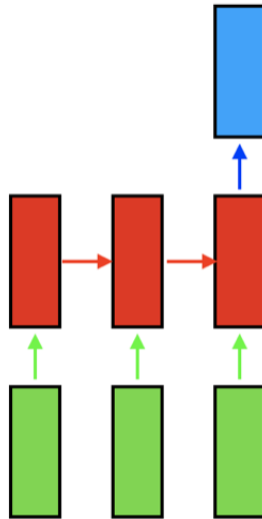
one to one



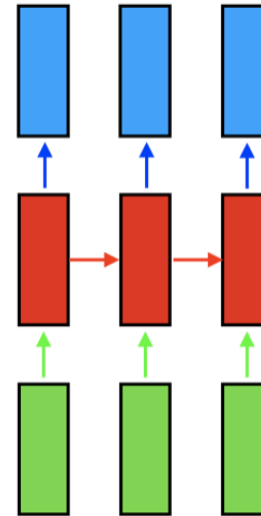
one to many



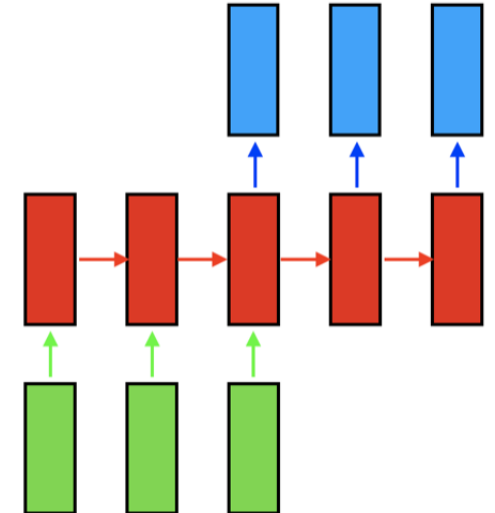
many to one



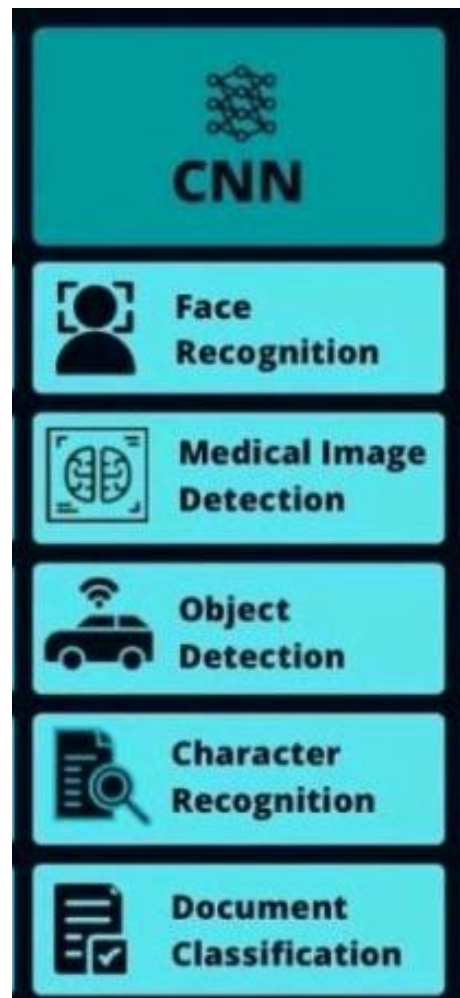
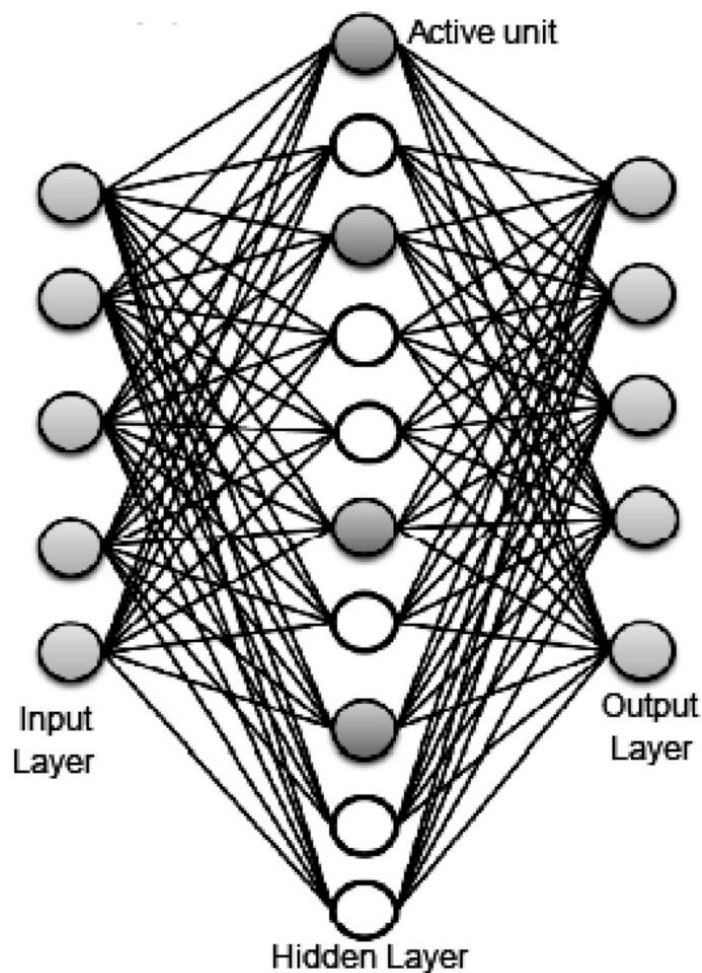
many to many



many to many

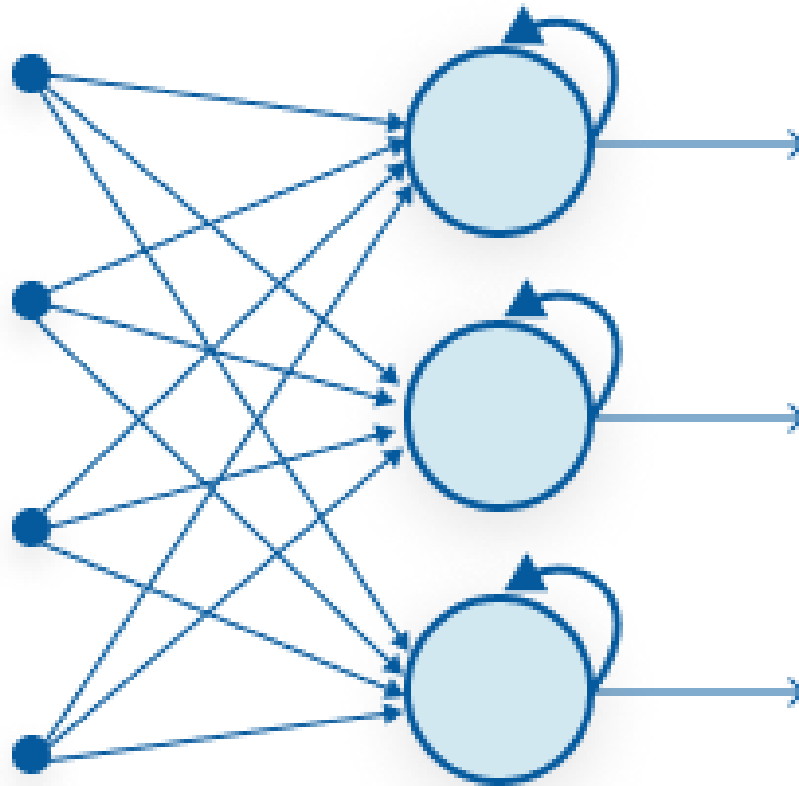


RNN

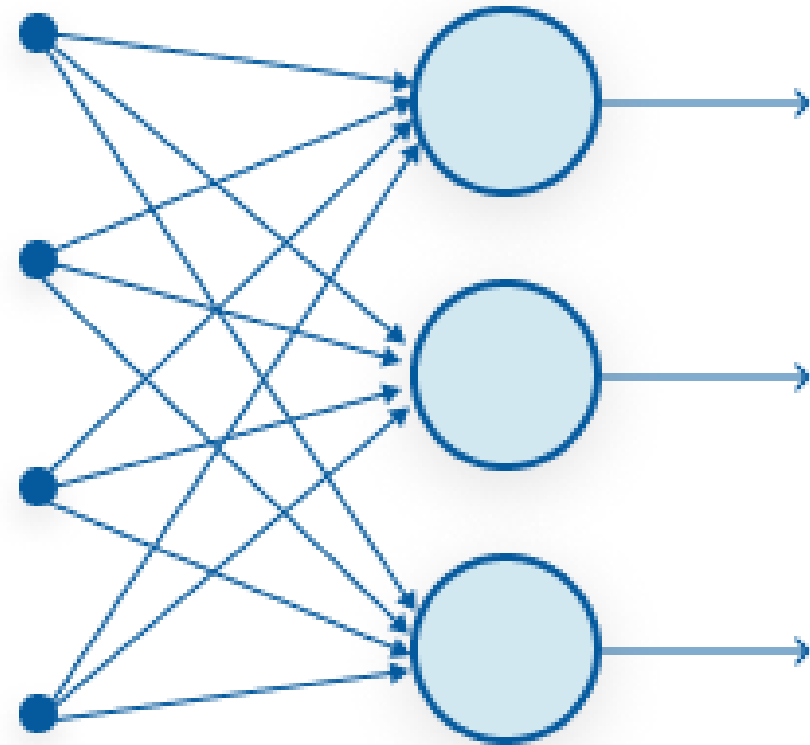


The background features abstract, flowing curved lines in shades of green and grey. A thick green line curves from the top left towards the center, while a grey line follows a similar path below it. On the right side, a large, thick green line forms a partial circle, with a grey line nested inside it.

DIFFERENCE BETWEEN RNN AND ANN



Recurrent Neural Network



Feed-Forward Neural Network

ANN



01

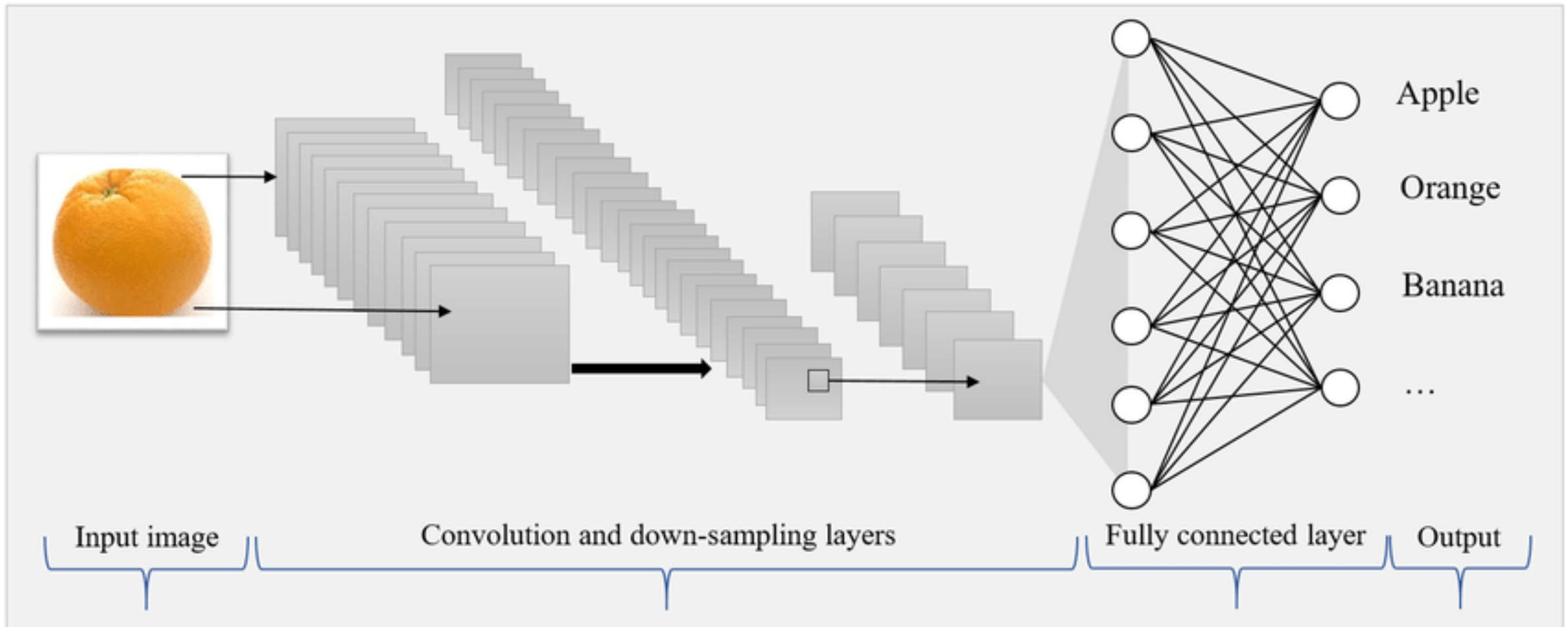
cannot handle
sequential data

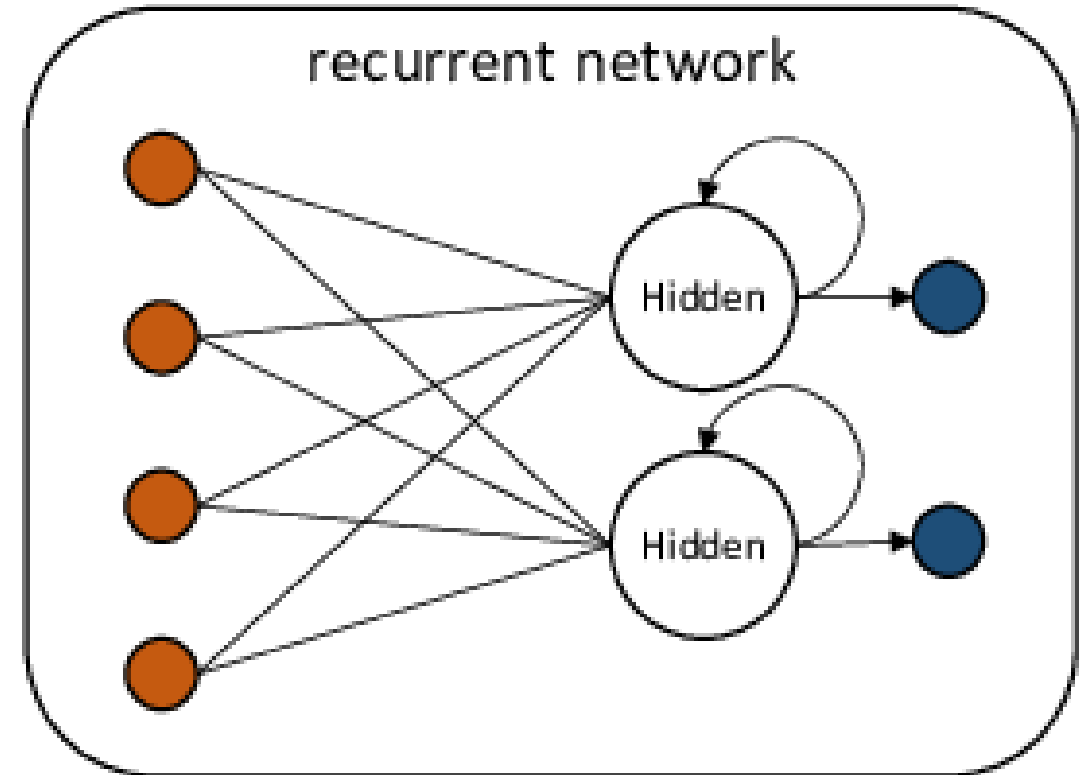
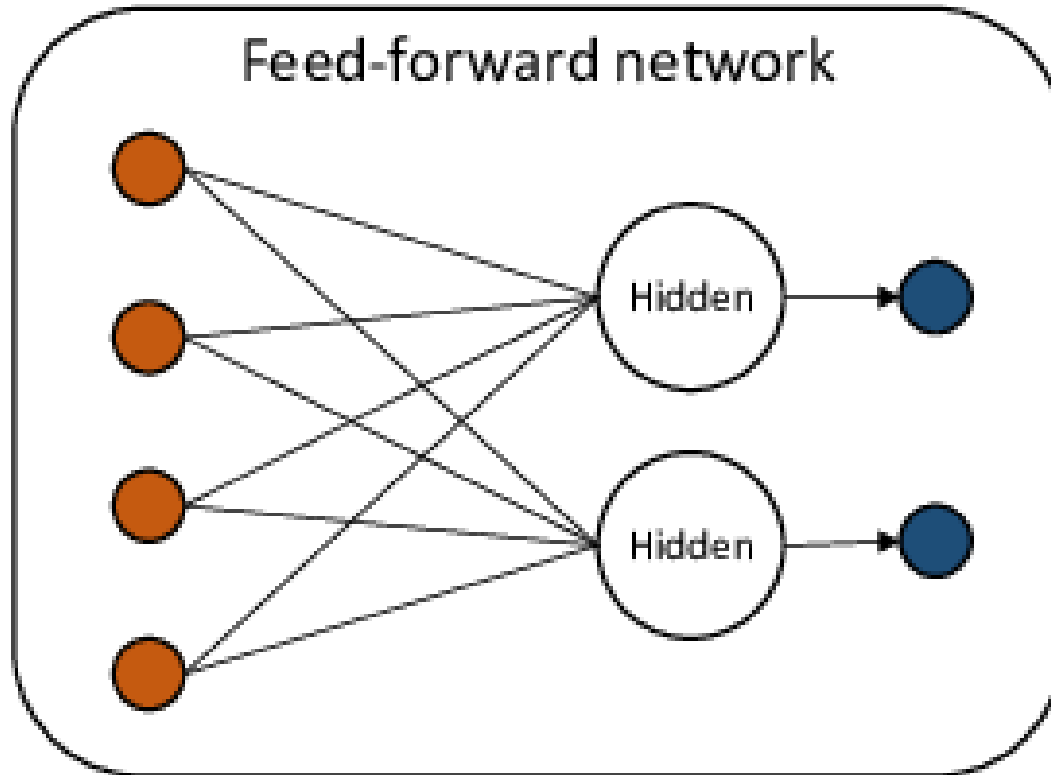
02

considers only the
current input

03

cannot memorize
previous inputs



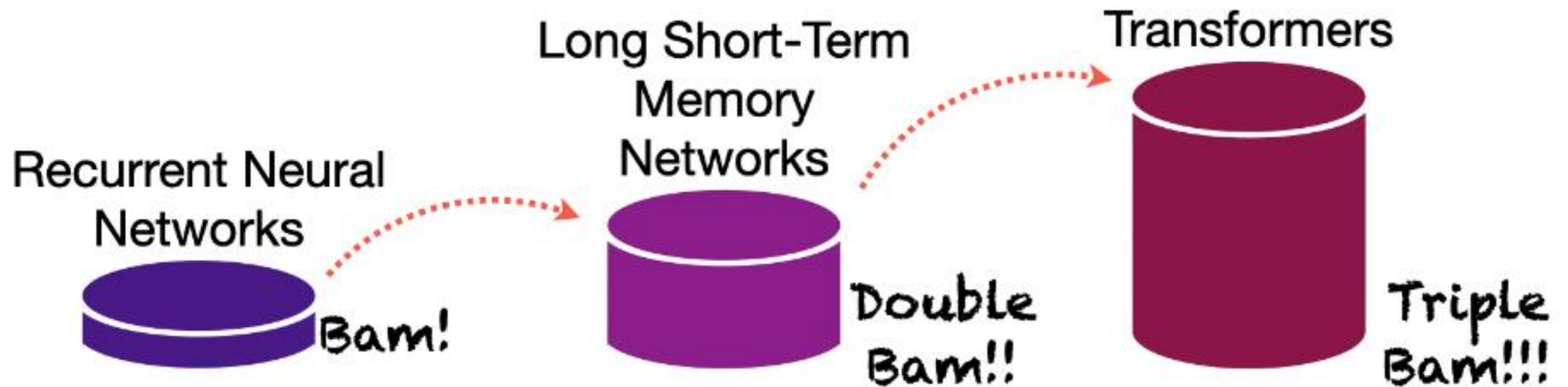


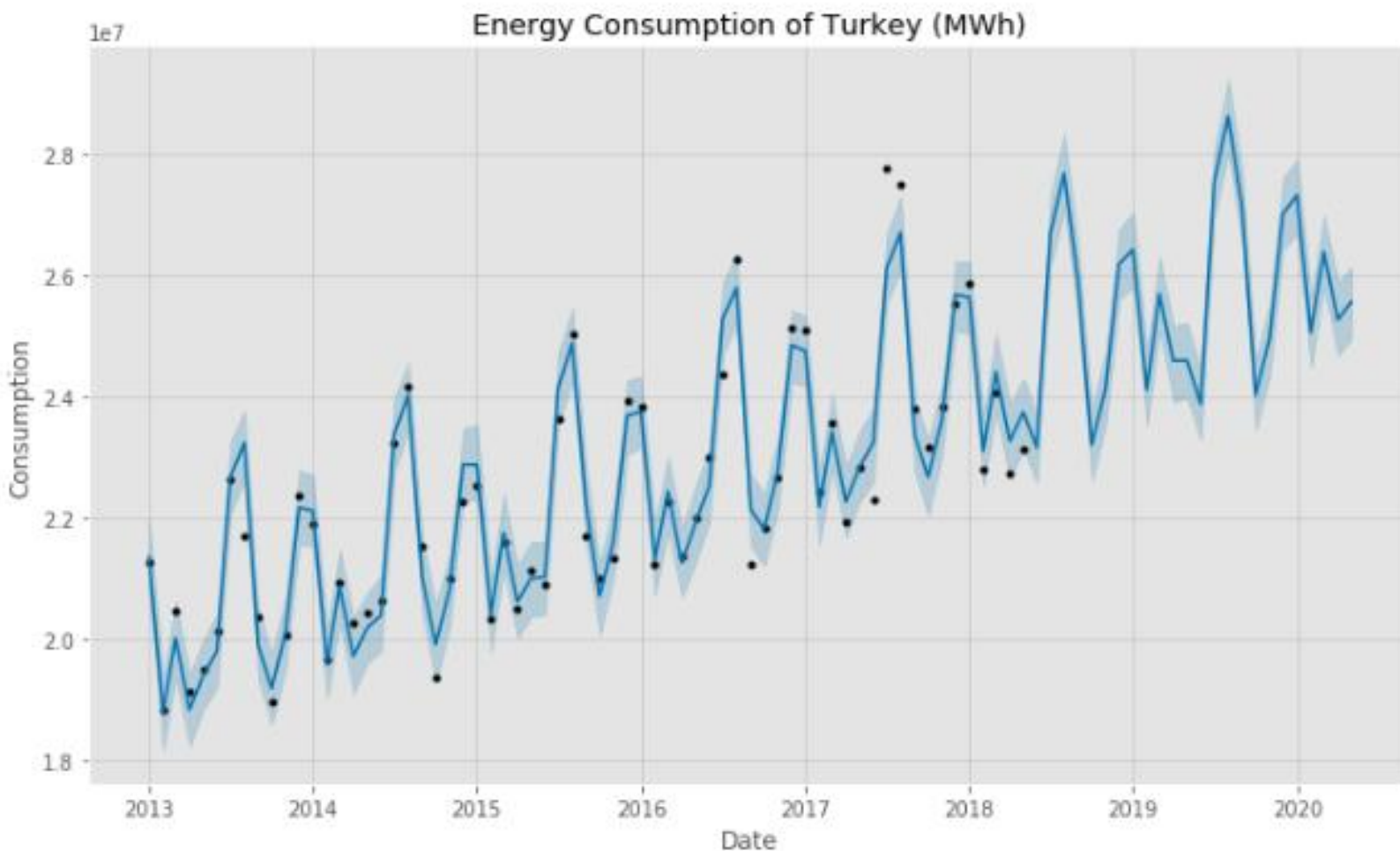
- ❖ RNNs are preferred in NLP because they are effective at modeling sequential data.
- ❖ The key difference between RNN models and ANN or CNN is that each layer in an RNN depends on the information from the previous layer. Without this information from the previous layer, it cannot function.
- ❖ In CNNs and ANNs, all data is processed independently of each other.
- ❖ Another key difference of RNN compared to CNN and ANN is that RNN has memory, which is why it is often preferred in NLP models.

Vanilla-RNN

**RNN (Recurrent Neural
Network)**







Time Series Example

	A	B
1	date	value
2	9/6/2017	531974.19
3	9/7/2017	484704.26
4	9/8/2017	693635.27
5	9/9/2017	420176.65
6	9/10/2017	257548.74
7	9/11/2017	212416.06
8	9/12/2017	410240.57
9	9/13/2017	559267.26
10	9/14/2017	556496.67
11	9/15/2017	813277.37
12	9/16/2017	600138.13
13	9/17/2017	371246.62
14	9/18/2017	319319.61
15	9/19/2017	561685.94
16	9/20/2017	650536.61
17	9/21/2017	599229.88

Time Series Example



THE USE CASES OF RNN IN REAL LIFE

RNN

RECURRENT NEURAL NETWORKS

SEQUENCE PROBLEMS

A Timeseries is a common example of this, with each point reflecting an observation at a certain point in time, such as a stock price or sensor data.

IN	OUT	PURPOSE
	THE QUICK BROWN FOX JUMPED...	SPEECH RECOGNITION
		MUSIC GENERATION
THERE IS NOTHING TO LIKE IN THIS MOVIE		SENTIMENT CLASSIFICATION
AGCCCCCTGTC AGGAACTAG	AGCCCCCTGTC AGGAACTAG	DNA SEQUENCE ANALYSIS
VOULEZ-VOUS chanter avec moi?	DO you want to sing with me?	MACHINE TRANSLATION
	RUNNING	VIDEO ACTIVITY RECOGNITION
Yesterday Harry Potter met Hermione Granger	Yesterday Harry Potter met Hermione Granger	NAME ENTITY RECOGNITION

Examples of sequence data

Speech recognition



^y
“The quick brown fox jumped
over the lazy dog.”

Music generation



Sentiment classification

“There is nothing to like
in this movie.”



DNA sequence analysis

AGCCCCTGTGAGGAACTAG



AG**CCCCTGTGAGGAACTAG**

Machine translation

Voulez-vous chanter avec
moi?



Do you want to sing with
me?

Video activity recognition



Running

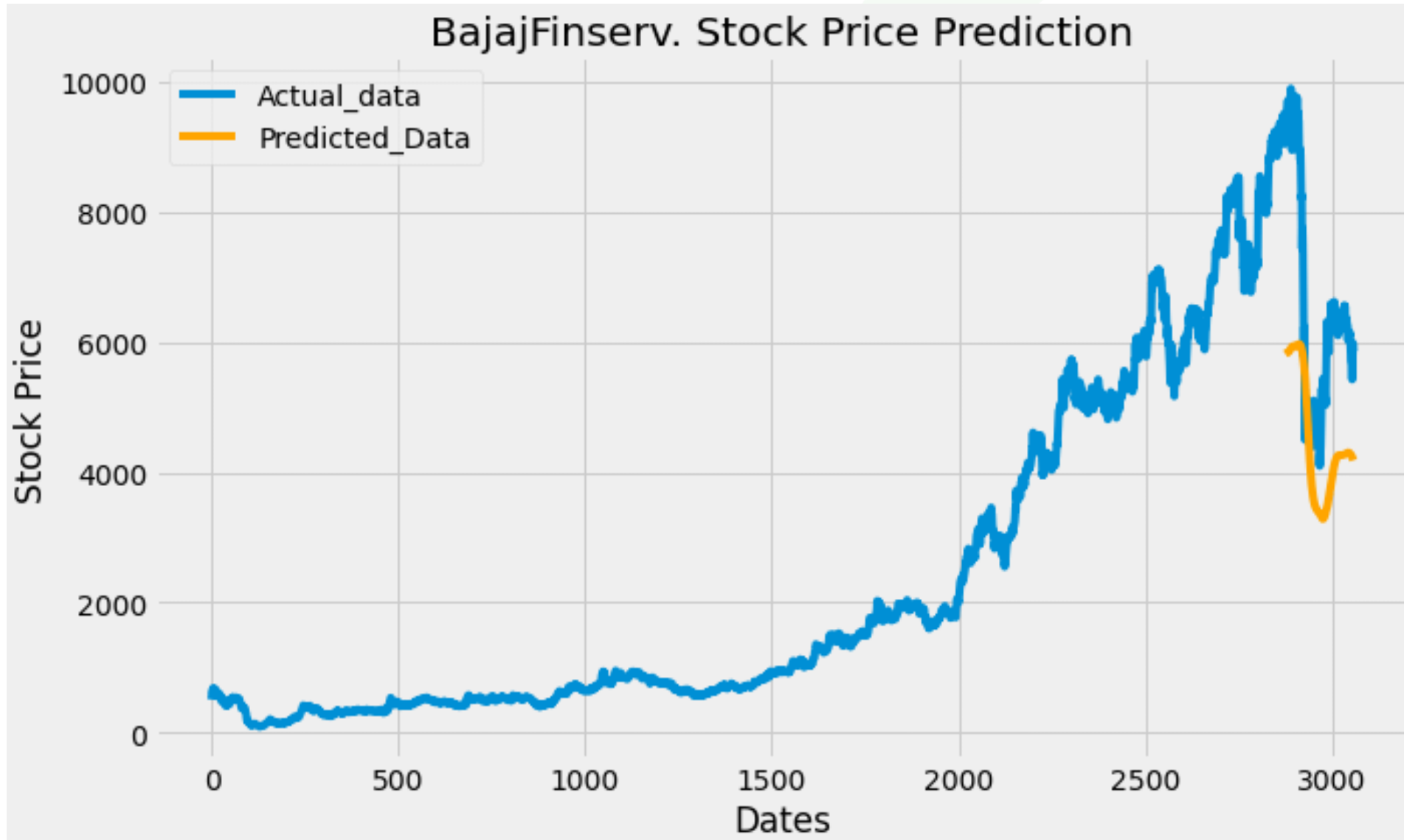
Name entity recognition

Yesterday, Harry Potter
met Hermione Granger.



Yesterday, **Harry Potter**
met **Hermione Granger**.

Andrew Ng



Text Mining and Sentiment Analysis can be carried out using RNN for Natural Language Processing

SENTIMENT ANALYSIS



RNN APPLICATION



Here the person is speaking in English and it is getting translated into Chinese, Italian, French, German and Spanish languages

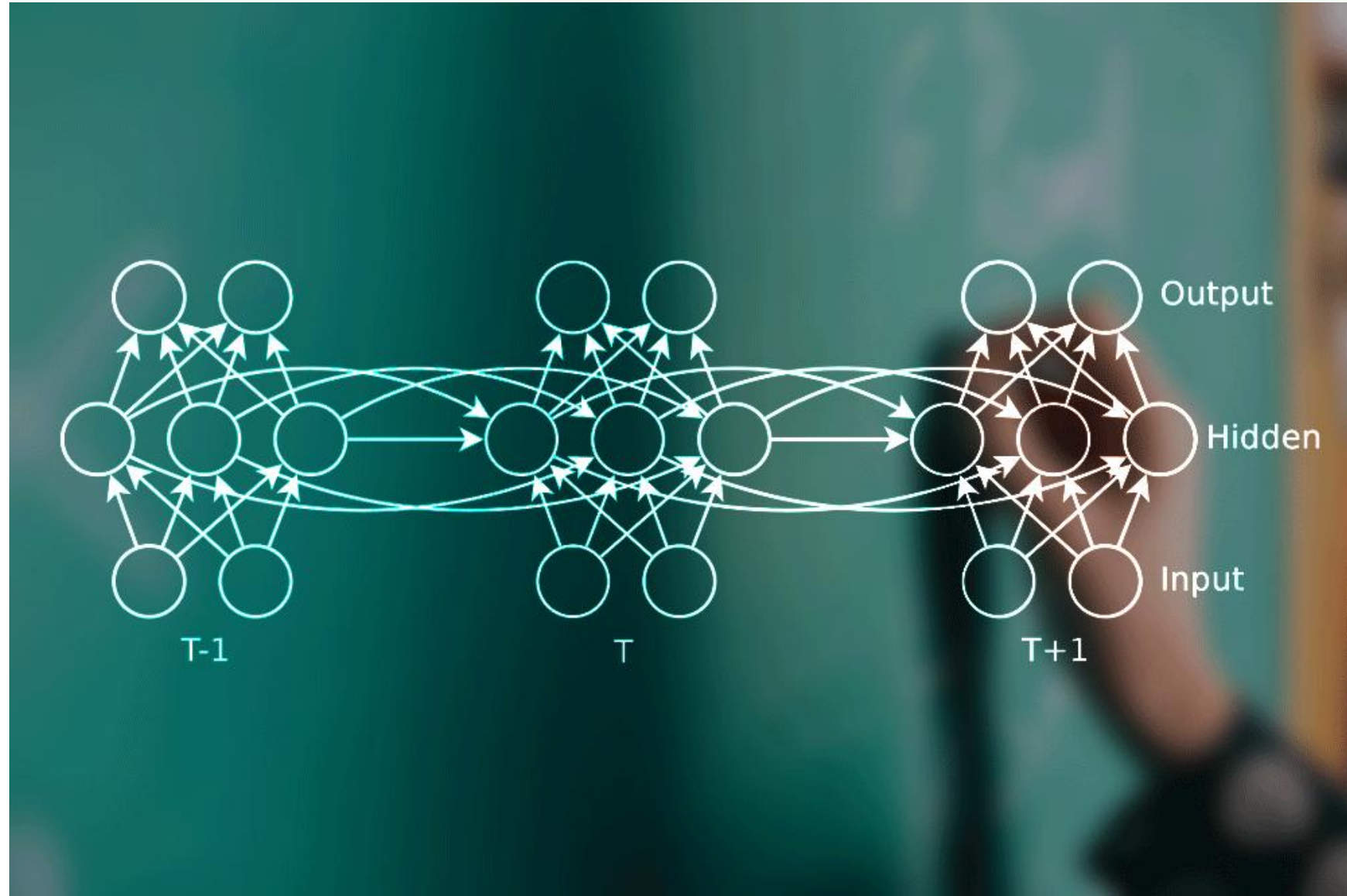
Machine Translation

Given an input in one language, RNN can be used to translate the input into different languages as output

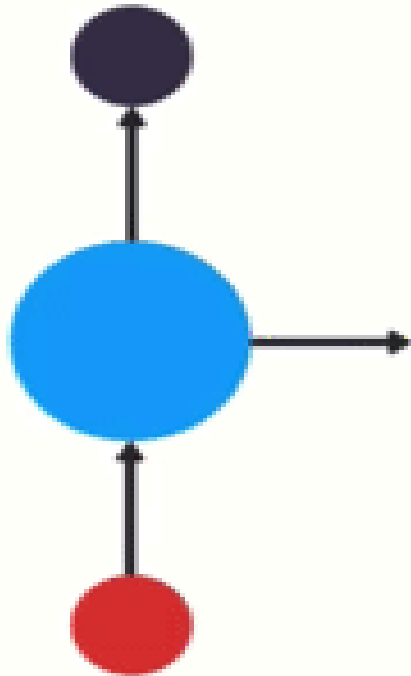
The background features two large, stylized, overlapping curved shapes. The shape on the left is a thick, greyish-green arc that curves from the top left towards the center. The shape on the right is a larger, similar arc that curves from the top right towards the bottom right. Both shapes have a slight gradient and a soft shadow effect.

HOW DOES RNN WORK?

A recurrent neural network (RNN) is a type of artificial neural network which uses sequential data or time series data.







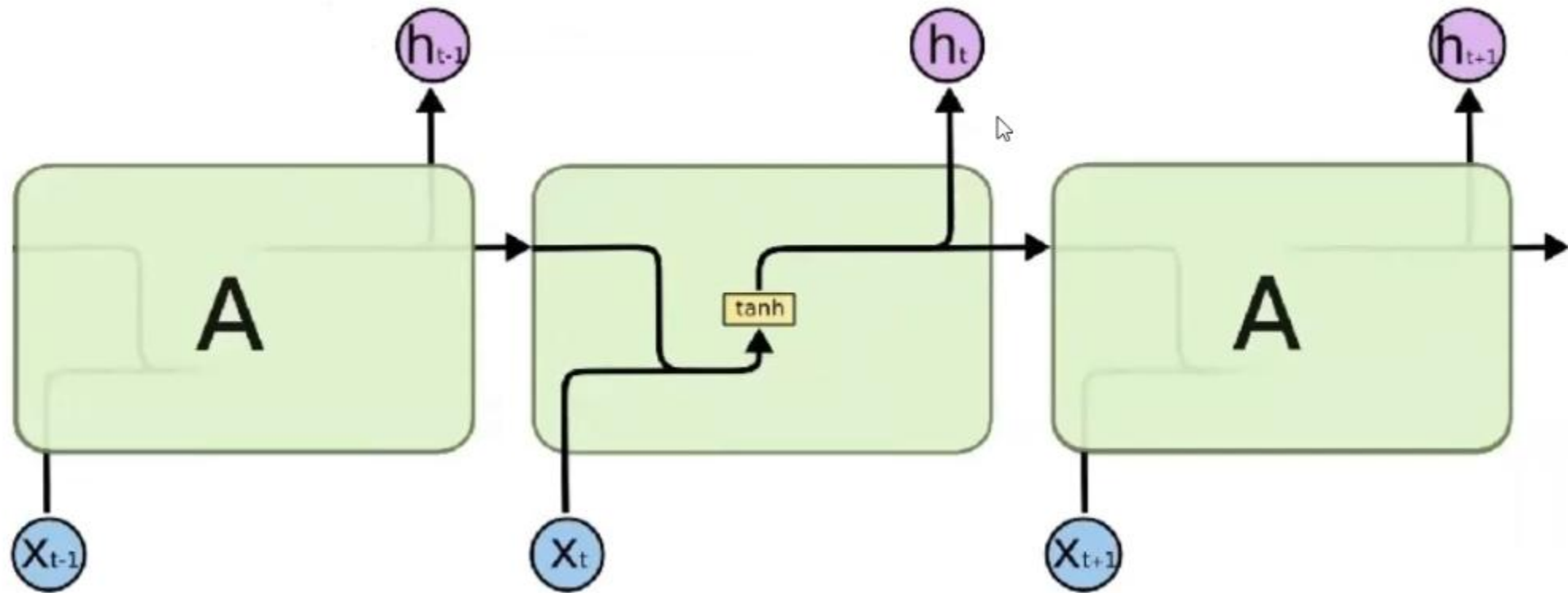
time_series_generator

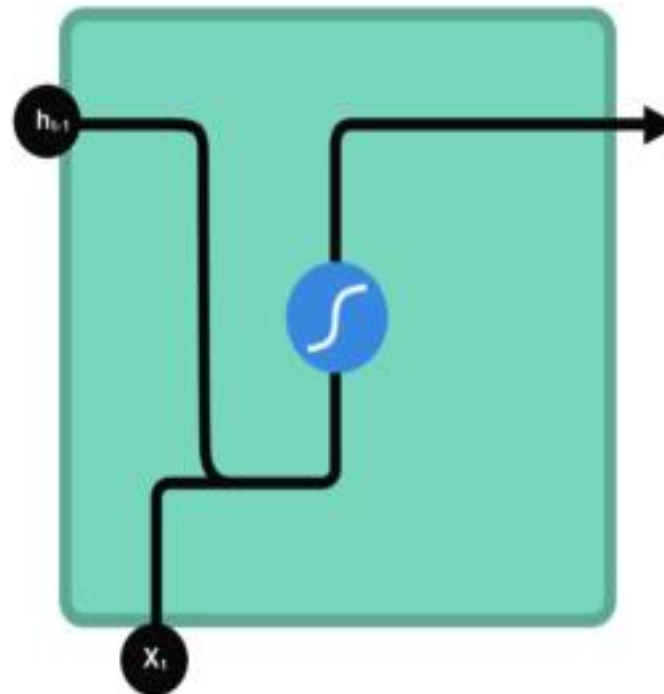
```
data = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

```
Grup 1 - Girdi (x): [[1 2 3]], Çıkış (y): [4]
Grup 2 - Girdi (x): [[2 3 4]], Çıkış (y): [5]
Grup 3 - Girdi (x): [[3 4 5]], Çıkış (y): [6]
Grup 4 - Girdi (x): [[4 5 6]], Çıkış (y): [7]
Grup 5 - Girdi (x): [[5 6 7]], Çıkış (y): [8]
Grup 6 - Girdi (x): [[6 7 8]], Çıkış (y): [9]
Grup 7 - Girdi (x): [[7 8 9]], Çıkış (y): [10]
```



```
batch_size = 1
```





Tanh function



new hidden state



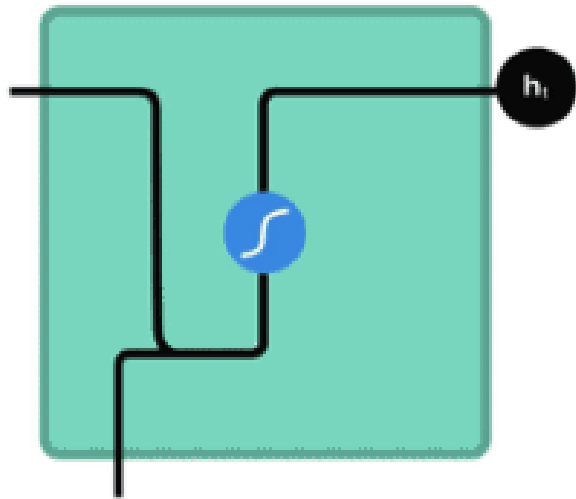
previous hidden state



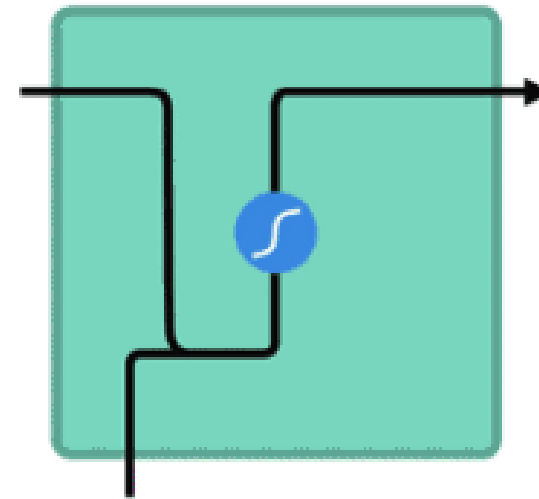
input



concatenation

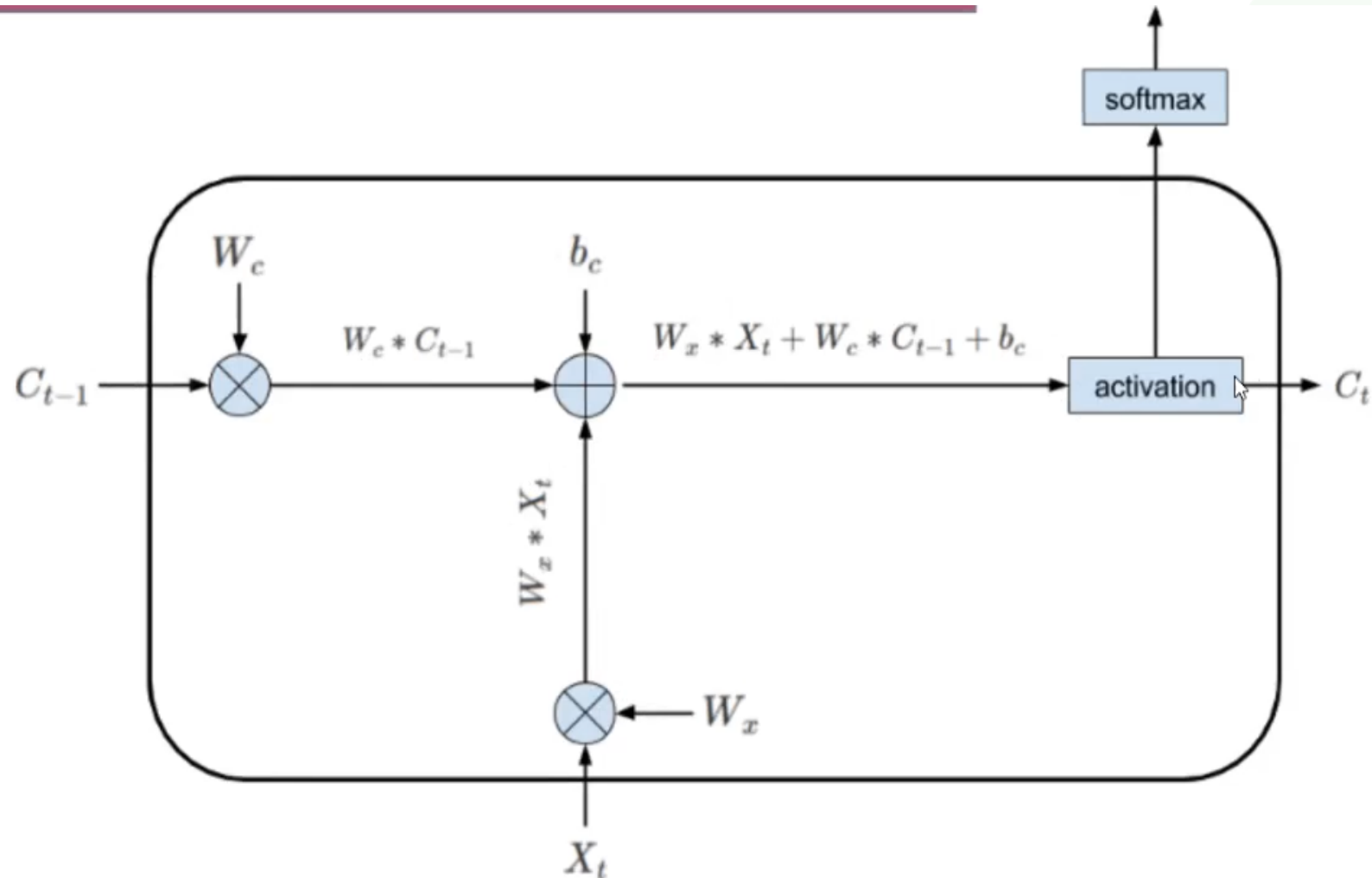


Tanh function



hidden state (memory)

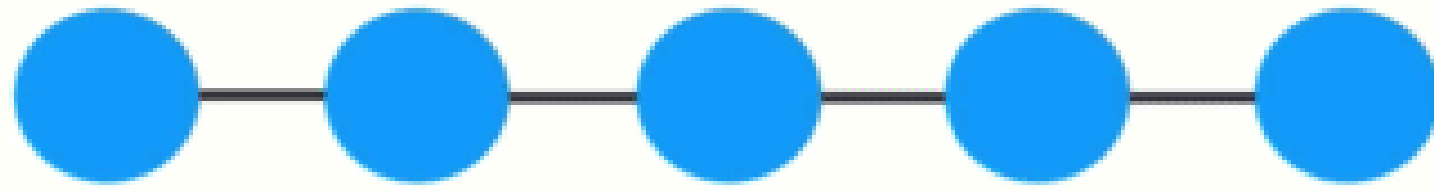
RNN



- X_t Is the input to the RNN at the time t
- C_{t-1} Is the cell state at time $t-1$
- W_c Is the weight matrix for C
- b_c Is the bias matrix
- W_x Is the weight matrix for X
- C_t Is the cell state at time t
- Y_t Is the output (effectively the prediction) of the RNN
-  Represents matrix multiplication
-  Represents matrix addition
- activation** Is the activation function (e.g. tanh activation would constrain values between 1 & -1)

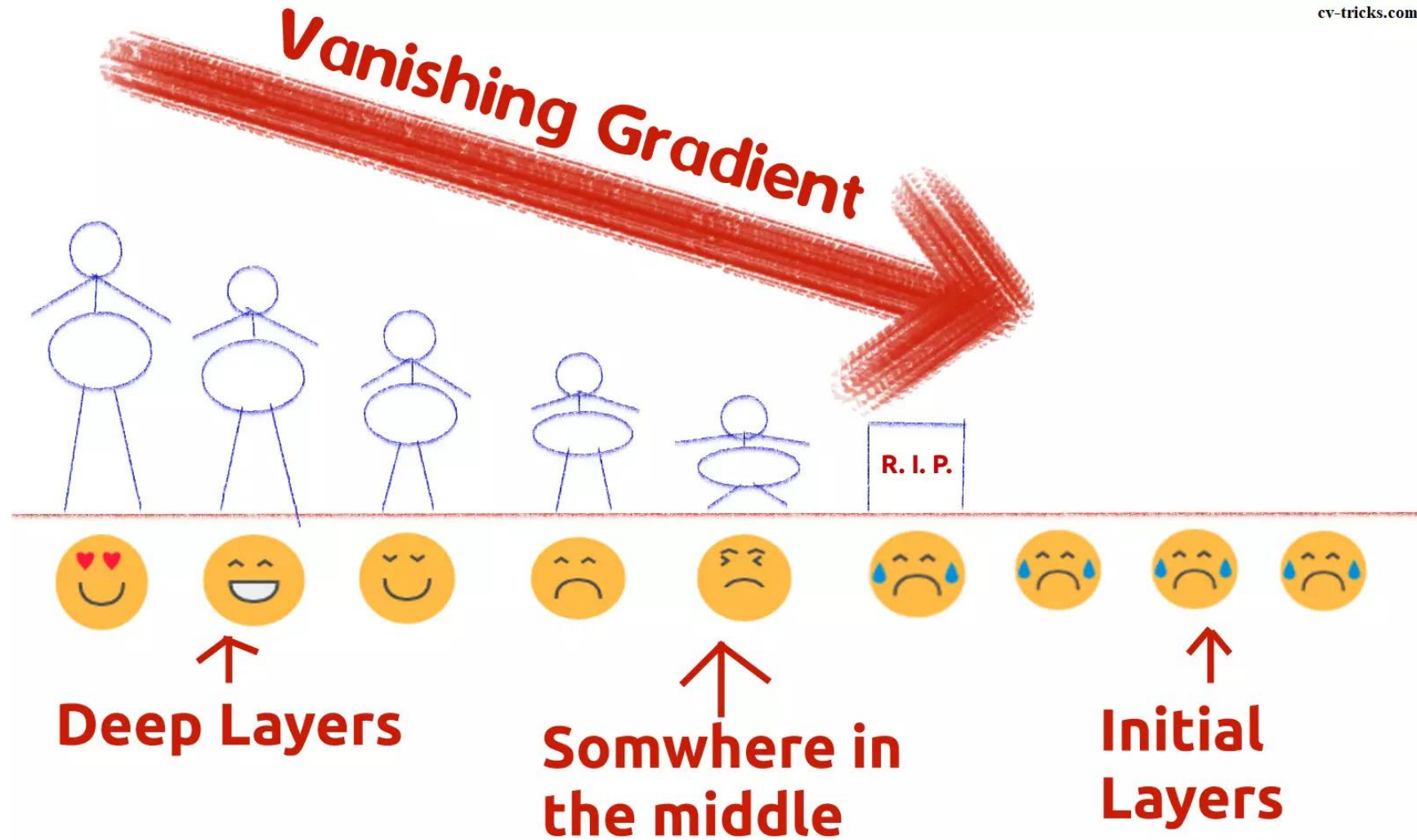


VANISHING /EXPLODING GRADIENT PROBLEM

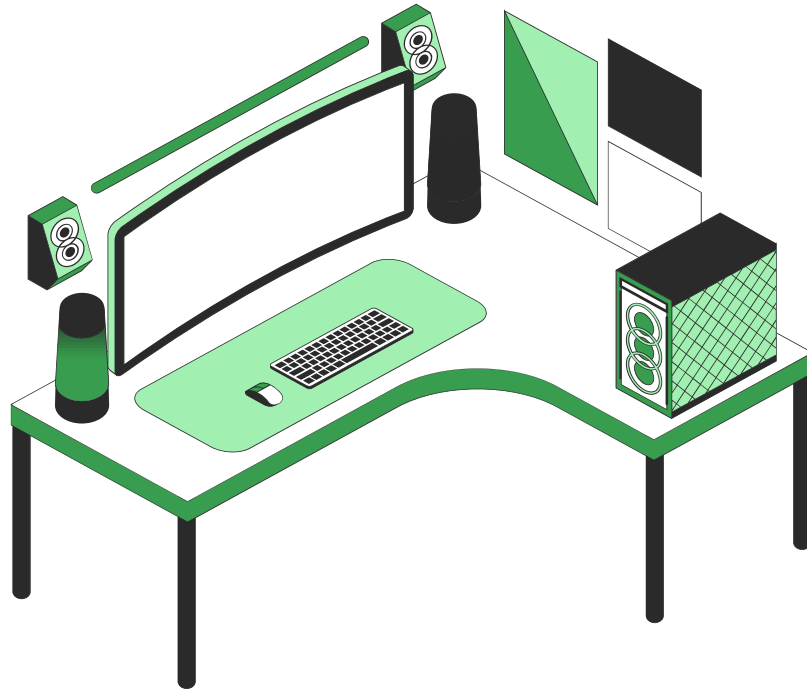


RNN VANISHING /EXPLODING GRADIENT

cv-tricks.com



THE END



Do you have any questions?

Send it to us! We hope you learned something new.

Tea break...

00:00



01

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